

# Optimizing Care Delivery in Street Medicine

Jeongsik Park, Joshua Blackmon, and Peter Phung

University of Southern California

{jeongsik, jlblackm}@usc.edu Peter.Phung@med.usc.edu

## Abstract

In an effort to explore ways to optimize and learn more surrounding the street care delivery within USC Keck’s Street Medicine Program in Los Angeles. Motivated by substantial variability in team productivity and rising demand for street-based care, we analyzed 3 months of encounter-level data to identify drivers of efficiency, characterize high-utilizer patients, and determine optimal daily scheduling targets. Using extensive feature engineering and predictive modeling, XGboost achieved strong performance ( $R^2 \approx 0.92$ ), and SHAP interpretability revealed that productivity is primarily driven by concentrated provider workload, higher true clinical time, and staffing compositions that are more dominated by MDs and PAs. Finally, a distributional robust linear program was developed to determine the minimum number of patients each team should schedule to reliably achieve six completed visits per day; results showed that RN capacity is not a binding constraint, while completion-rate uncertainty became the limiting factor. The study provides a data-driven foundation for staffing exploration and targeted patient engagement strategies that support more consistent, efficient care delivery in street medicine operations.

## 1 Introduction

Los Angeles County has one of the largest unhoused populations in the U.S., and traditional healthcare systems often fail to reach people experiencing homelessness (PEH) due to barriers like a lack of transportation and variable health needs (Feldman et al., 2021). Street medicine serves to meet the needs of PEH by combining medical, behavioral, and social services in low-barrier, community-based settings. Street medicine is specifically defined as the direct delivery of healthcare in encampments, on sidewalks, and in shelters (Medellin et al., 2024). Kaufman et al. (2024) shows that street medicine and mobile clinics significantly improve access to primary and

behavioral healthcare, support housing and insurance enrollment and serve to strengthen trust between providers and unhoused patients, enhancing engagement and continuity of care.

Keck Medicine’s Street Medicine Program provides a direct response by meeting patients where they live, on sidewalks, in encampments, in cars, and in shelters, rather than requiring clinic attendance. The program reports reduction in hospitalizations, reduction in hospital length of stay, and increase in housed population among regular program participants. The clinical model relies on multidisciplinary teams consisting of physicians, advanced practice providers, nurses, social workers, and community health workers. Clinical activities range from preventive care to chronic disease management, mental health treatment, wound care, reproductive health, and substance use disorder treatment. Despite the program’s impact, operational inefficiencies persist, driven by variability in staffing, travel distances, high no-show rates, nonuniform provider skill sets, and highly heterogeneous patient needs. Understanding how staffing structure influences productivity, how early utilization patterns can predict high-need patients, and how to determine robust scheduling targets is essential to improving care delivery.

Our team spoke with the Keck Medicine Street Medicine leadership team to better understand their operational challenges. Street Medicine leadership identified four central operational questions:

1. Why are some teams more productive than others?
2. How does staffing mix—especially RN staffing—affect provider workload and outcomes?
3. How can we identify high-utilizer patients early?
4. How many patients should each team schedule per day to reliably achieve six completed visits?

Each question corresponds to a real operational challenge:

- Productivity variation creates inequity in patient access.
- Staffing variability affects encounter rates and provider burnout.
- Early identification of high utilizers allows proactive care planning.
- Effective scheduling reduces downtime and inefficient routing.

The goal of this project is to use data-driven approaches like feature engineering, supervised machine learning, SHAP interpretability, and robust linear optimization to provide evidence-based operational recommendations.

## 2 Related Work

Lynch et al. (2022) highlight the financial and operational challenges of sustaining street medicine. They estimate annual costs between \$3,000–\$9,000 per patient but demonstrate potential savings through reduced emergency utilization. They propose innovative funding mechanisms, including shared savings and global capitation models, and emphasize the need for rigorous data collection and predictive evaluation to guide efficient resource allocation. Building on this gap, this project develops predictive and optimization models to determine where and when care should be delivered, forecast healthcare demand, and improve the reach, efficiency, and equity of street medicine delivery in Los Angeles.

A recent meta-narrative review synthesized real-world applications of AI and machine learning for patient scheduling, demonstrating modest improvements in no-show prediction, overbooking, and resource allocation, primarily within traditional outpatient and operating room settings (Knight et al., 2023). In contrast, our study applies data-driven scheduling methods to a Street Medicine program, addressing fundamentally different operational constraints such as mobile care delivery, housing instability, and high rates of missed encounters.

## 3 Dataset

### 3.1 Dataset Source

We analyzed routinely collected clinical and administrative data from the USC Street Medicine Program in Los Angeles, covering all scheduled patient encounters from July to September 2025. The

dataset integrates patient-level demographic information (age, gender, race/ethnicity, language, and insurance status) with encounter-level operational data, including visit dates, locations, and clinician identifiers. Unique patient identifiers enable longitudinal tracking across visits, and no-show encounters were explicitly recorded. Detailed clinical notes were not available in this dataset.

**Data Cleaning and Integration** Raw encounter-level data were collected from three monthly spreadsheets spanning July to September 2025. Prior to integration, all datasets were standardized to ensure schema consistency across files, and records with inconsistent or invalid entries were removed. The cleaned datasets were then concatenated into a single unified dataframe comprising 3,928 team-day observations, which served as the input for all subsequent modeling and analysis. This preprocessing step ensured a consistent feature space across time and minimized noise prior to predictive modeling. A detailed description of all variables used in the analysis is provided in Appendix A.

### 3.2 Appointment-Level Feature

Following clinical guidance, we define providers as MD/DO, NP, PA, RN and treat CHW as a non-provider whose encounters serve as a no-show analog rather than billable clinical visits. We implement role standardization by enumerating unique RESOURCE–role pairs, assigning each RESOURCE to a canonical role (e.g., HINTON PA, ANNE→PA), and merging the mapping back so every appointment row has a standardized role.

We split appointment duration into provider-facing clinical time and CHW/no-show-analog time:

$$\begin{aligned} \text{duration\_normal} &= \text{APPT\_DURATION}(1 - \text{is\_chw}), \\ \text{duration\_noshow} &= \text{APPT\_DURATION} \cdot \text{is\_chw}. \end{aligned}$$

We then aggregate to the date–team–role level  $(d, t, r)$  and compute:

- count (encounter count)
- $\sum \text{duration\_normal}$
- $\sum \text{duration\_noshow}$
- unique\_providers (number of distinct RESOURCE in the group)

From these we derive per-provider role-level measures when  $\text{unique\_providers} > 0$ :

- $\text{count\_per\_provider} = \frac{\text{count}}{\text{unique\_providers}}$
- $\text{duration\_normal\_per\_provider} = \frac{\text{duration\_normal}}{\text{unique\_providers}}$

### 3.3 Team-Day Aggregation and Productivity Definition

We aggregate further to team-day  $(d, t)$ , tracking both overall workload and provider-only workload (MD/DO, NP, PA, RN). We define totals:  $\text{total\_encounters}$ ,  $\text{total\_duration\_normal}$ , and  $\text{total\_duration\_noshow}$ , and provider-only quantities:  $\text{provider\_encounters}$ ,  $\text{provider\_duration}$ , and  $\text{provider\_unique}$ .

### 3.4 Role Mix and Provider Composition

To characterize staffing mix, we pivot team-day role encounter totals and collapse providers into three groups: (1) MD (physicians), (2) NP+PA (advanced practice providers), and (3) RN (registered nurses). Let MD, NP+PA, and RN denote their encounter totals on a team-day. We define total provider volume:

$$\text{provider\_total\_by\_role} = \text{MD} + \text{NP+PA} + \text{RN}.$$

We then compute composition shares (set to 0 when the denominator is 0):

$$\begin{aligned} \text{MD\_share\_among\_providers} &= \frac{\text{MD}}{\text{provider\_total\_by\_role}}, \\ \text{NP+PA\_share\_among\_providers} &= \frac{\text{NP+PA}}{\text{provider\_total\_by\_role}}, \\ \text{RN\_share\_among\_providers} &= \frac{\text{RN}}{\text{provider\_total\_by\_role}}. \end{aligned}$$

These variables capture the team's daily *skill mix* as observed through role-specific encounter volumes.

### 3.5 Outcome Definitions

We define three primary outcomes aligned with the study objectives:

1. Per-Provider productivity, defined as:

$$\text{Productivity} = \frac{\text{Provider Encounters}}{\# \text{ of Unique Providers}}$$

This is the average number of provider-level encounters completed per distinct provider on a team-day. For example, if a team performs 20 provider encounters with 4 distinct providers (1 MD, 2 NP, 1 PA), then  $\text{provider\_prod\_per\_provider} = 20/4 = 5$ . Notably, since productivity is defined

per provider, increases in provider headcount mechanically reduce the outcome unless encounter volume grows proportionally.

2. High-Utilizer Patient Label, binary indicator that determines whether a patient was a high-user, client defined as:

$$\text{Number of Visits} \geq 6$$

$$; \text{Number of Provider Visits} \geq 4$$

3. Completion Likelihood with a normal-approximation lower confidence bound, used to model the probability that a scheduled visit is completed, informing our later models. Defined as:

$$p_{low,t} = \hat{p}_t - 1.96 \sqrt{\frac{\hat{p}_t(1 - \hat{p}_t)}{N_t}}$$

### 3.6 Modeling Dataset and Leakage Control

For predictive modeling, the target variable is per-provider daily productivity. To reduce information leakage and redundancy, we exclude identifiers, the outcome and its direct derivatives, and raw role counts summarized by composition variables. Team labels are one-hot encoded, producing the final feature matrix  $X$ .

These choices were all made with the intention of applying them to a larger dataset that spans around a year, if not years. After extensive cleaning and feature engineering, the dataset supports predictive modeling, team productivity analysis, and distributionally robust scheduling optimization. Finally, further simplicity and conservative estimates were found with mindfulness of our clients in mind.

## 4 Methods

Our analysis integrates predictive modeling, model interpretation, and robust optimization to address key operational questions in street medicine. We model per-provider daily productivity using multiple regression approaches, interpret the selected model using SHAP values. Also, we construct a distributionally robust optimization model to determine scheduling targets under completion uncertainty and RN capacity constraints.

## 4.1 Predictive Modeling

We evaluate three regression models: Lasso regression (LassoCV) (Tibshirani, 1996), Random Forest (Breiman, 2001), and XGBoost (Chen and Guestrin, 2016a), using  $R^2$  and RMSE as performance metrics.

**Cross-Validation** We perform 5-fold cross-validation with shuffling (random\_state=42). Performance is evaluated using  $R^2$  and RMSE (encounters per provider per day).

**Model Selection** Lasso is trained using LassoCV with an inner cross-validation for regularization. Random Forest and XGBoost use fixed hyperparameters (RF: 300 trees; XGB: 400 trees, depth 3, learning rate 0.05, subsample 0.8, colsample\_bytree=0.8). Across folds, XGBoost achieves the strongest predictive accuracy and is therefore selected as the final model.

**Final Model Training** The selected XGBoost model is trained on the full dataset  $(X, y)$ .

## 4.2 Model Interpretation (SHAP)

To interpret the predictions of the final model, we apply SHAP (Shapley Additive Explanations) (Lundberg and Lee, 2017). SHAP values are computed using TreeExplainer to quantify each feature’s contribution to predicted productivity at the team-day level.

We generate (i) a global SHAP summary plot to rank features by mean absolute contribution and (ii) SHAP dependence plots for key variables. Results are presented in Section 5.

## 4.3 Robust Scheduling Optimization

To determine scheduling targets under completion uncertainty, we formulate a distributionally robust linear program that incorporates lower-bound completion probabilities and RN capacity constraints.

$$\min_{n_t} n_t \quad (1)$$

$$\text{s.t. } n_t \cdot p_{\text{low},t} \geq 6, \quad (2)$$

$$n_t \leq n_{\text{max},t}, \quad (3)$$

$$a_t n_t \leq \text{RNcap}_t, \quad (4)$$

$$n_t \in \mathbb{Z}_+, \quad (5)$$

where:

- $n_t$  = number of scheduled patients for team  $t$ ,
- $p_{\text{low},t}$  = lower-confidence bound of completion probability for team  $t$ ,
- $n_{\text{max},t}$  = maximum feasible number of scheduled visits based on historical capacity,
- $a_t$  = average RN time required per scheduled encounter,
- $\text{RNcap}_t$  = total RN time capacity available to team  $t$  per day.

Constraint (2) ensures each team achieves at least six completed visits even under worst-case completion rates. Constraint (3) enforces operational scheduling limits. Constraint (4) restricts schedules based on RN time availability; dual analysis later confirms this constraint is slack. Constraint (5) ensures decision variables are non-negative integers.

## 5 Results & Discussions

### 5.1 RQ1: Why Are Some Teams More Productive Than Others?

**Sanity checks: workload volume and clinical time.** In Figure 1, the SHAP summary plot indicates that overall provider workload volume and provider clinical time are among the strongest drivers of per-provider productivity. Both provider\_total\_by\_role and total\_duration\_normal exhibit monotonic positive SHAP patterns: team-days with more provider encounters and more true clinical time yield higher predicted productivity. This validates the productivity definition by aligning model behavior with operational intuition.

**Unique provider headcount: concentration vs. dilution.** In Figure 1b, provider\_unique shows a nonlinear relationship with productivity. SHAP dependence plots suggest that days staffed by 1–2 distinct providers tend to have higher per-provider productivity, while productivity declines as the number of distinct providers increases (e.g., 4–6 providers). This pattern is consistent with a dilution effect: encounters are spread across more individuals, lowering average encounters per provider.

### 5.2 RQ2: How Does Staffing Mix Affect Outcomes and Provider Workload?

Increasing provider\_unique is associated with lower per-provider productivity, implying that

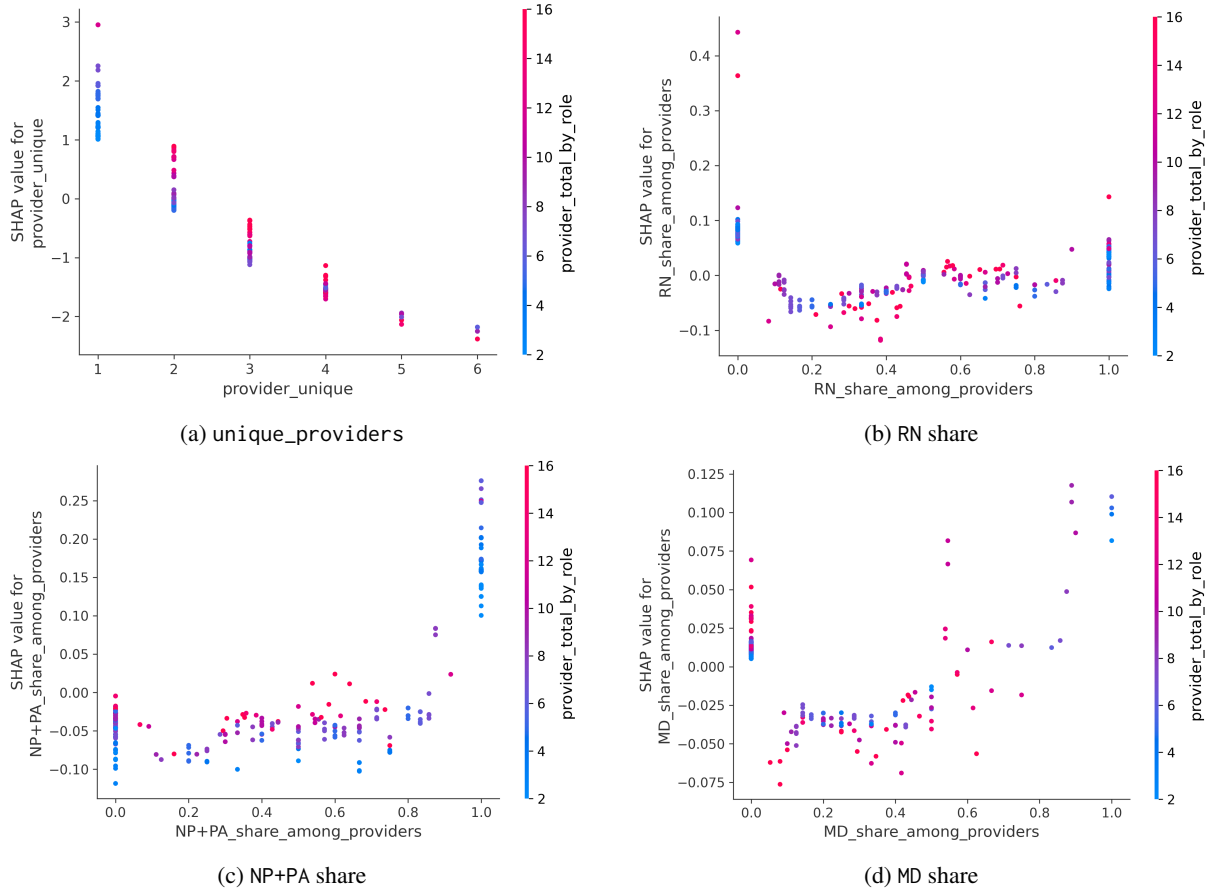


Figure 1: Staffing Mix. The left y-axis shows the SHAP value, indicating how strongly the x-axis variable is positively or negatively related to productivity. The right y-axis shows the total number of provider encounters for the team on that date. (a) unique\_providers. (b) Share of RNs among providers. (c) Share of NPs and PAs among providers. (d) Share of MDs among providers.

adding distinct providers often diffuses workload rather than increasing per-provider throughput. To identify which provider-dominant staffing mix increases productivity, we analyzed productivity for each provider share.

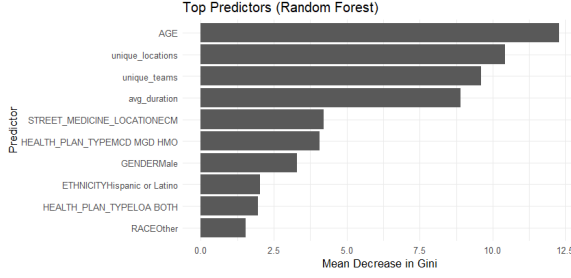
**RN share: supportive but not sufficient.** Figure 1b exhibits a nonlinear effect. A moderate RN share is weakly positive, consistent with a supportive role, while a very high RN share becomes negative, indicating that RN-dominant days lack sufficient MD/NP+PA capacity to sustain provider-level throughput.

**NP+PA and MD share: efficient provider capacity.** Higher NP+PA\_share\_among\_providers in Figure 1c and MD\_share\_among\_providers in Figure 1d are associated with higher predicted productivity. These patterns suggest that productivity is highest when a small number of high-capacity providers (MD and NP/PA) carry a larger share of provider-level encounters, complemented by RN support.

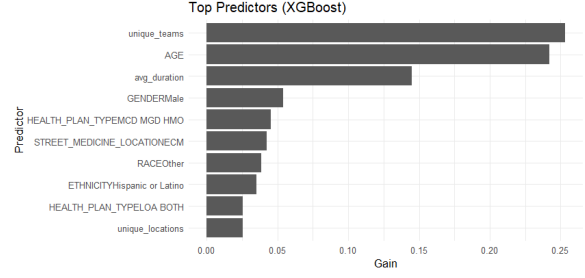
### 5.3 RQ3: How can we identify high utilizer patients early?

**Different Teams, Locations, Age, and Duration are the largest Indicators** Using similar methods as before, we aimed to use the models to predict high utilized patients and then assessed variable importance on each model’s importance metric natively. Random Forest and XGBoost were chosen because they capture nonlinearities and interactions while providing interpretable variable importance, and agreement between bagging- and boosting-based ensembles increases robustness against model-specific artifacts (Breiman, 2001; Chen and Guestrin, 2016b). Across both models, you can see in Figure 2 how important unique teams/locations, age, and average duration are to each model. What this effectively means is that older patients and patients that take longer to process through the system use the services more. One of the more interesting results of this as well is that patients that are seen by different teams (as in dif-





(a) Top Predictors for Random Forest



(b) Top Predictors for XGBoost

Figure 2: Comparison of Top Predictors Across Tree-Based Models

| Team      | $p_{low}$ | Mean RN Count | $n_{LP,RN}^*$ |
|-----------|-----------|---------------|---------------|
| CD1       | 0.945     | 11.20         | 7             |
| Eastside  | 0.715     | 7.43          | 9             |
| Hollywood | 0.678     | 7.18          | 9             |
| Mid City  | 0.838     | 8.41          | 8             |
| Other     | 0.945     | 36.50         | 7             |
| South LA  | 0.432     | 2.38          | 14            |

Table 1: Team-Level RN Staffing and Robust LP Results

ferent teams that go to the same area) or are seen at different locations (as in patients that move around team’s service areas) tend to be higher utilizers. This could be telling of how dismantled each team is or how patients in these situations tend to be more complicated.

#### 5.4 RQ4: How many patients should each team schedule per day to reliably achieve 6 completed visits?

**Each Patient Count is dependent on Show-up Rate** Here is when we use an Integer Program to optimize the best way to schedule patients as seen in Equations 2 through 4. In Table 1, it can be seen how there’s wide variation between each team’s show-up rate (defined as  $p_{low}$  with rates between 0.40 and 0.95 thus deriving the answer of our question in  $n_{LP,RN}^*$ . Notably, the inclusion of RN capacity doesn’t change the efficiency of the work denoting that it is not the bottleneck to the operation under current probabilities supporting the answer of Question 2.

#### 5.5 Operational Implications

Overall, results suggest that (i) concentrated provider rosters (few distinct providers) are associated with higher per-provider productivity, (ii) balanced skill mix matters, with MD and NP/PA-dominant days generally yielding higher throughput than RN-dominant days, (iii) CHW activity

captures a distinct operational signal that is negatively associated with provider-level productivity under our metric, and (iv) the highest utilizers tend to be older and seen by a high variety of teams.

#### 5.6 Discussions

This study demonstrates the feasibility of applying machine learning and optimization methods to address core operational challenges in Street Medicine, a care model characterized by mobility, uncertainty, and highly heterogeneous patient needs. By combining productivity modeling, staffing mix analysis, early identification of high utilizers, and robust scheduling optimization, this work illustrates how data-driven operational methods can be applied in a setting where traditional clinic-based assumptions do not hold.

Across analyses, team productivity varied substantially, suggesting that staffing structure and scheduling practices may meaningfully influence daily output in Keck Medicine’s Street Medicine Program. Rather than drawing conclusions about any single driver of productivity, these findings underscore that there may be opportunities to improve care delivery through staffing and scheduling decisions, and motivate further study to understand which approaches are most appropriate in Street Medicine programs.

Staffing composition appeared to influence how teams convert scheduled encounters into completed provider visits. While registered nurses play an essential role in patient education, safety, and longitudinal care, RN availability was not a binding constraint in achieving daily visit targets under current workflows. This finding should be interpreted cautiously, as encounter-based productivity measures do not capture downstream clinical benefits or care quality, particularly for nursing-led interventions.

The ability to identify high-utilizer patients early

in their care trajectory offers an opportunity for more proactive resource allocation. In Street Medicine programs, where a small subset of patients often accounts for a large share of encounters, early recognition may support more intentional care planning and help prevent escalation to emergency department visits or acute hospital admissions.

Finally, the scheduling optimization results underscore that uncertainty in visit completion is a defining operational challenge in Street Medicine. By explicitly incorporating completion variability into scheduling targets, the robust optimization approach demonstrates how teams can plan for missed encounters while still achieving consistent service delivery, reframing no-shows as an expected feature rather than an operational failure.

## 6 Conclusion

This study is subject to several limitations that motivate future extensions. First, the available data lacked specific temporal and spatial detail (routing information, travel time between encounters, and intra-day sequencing of visits). Incorporating richer spatiotemporal data would allow more accurate modeling of operational constraints and productivity drivers.

Second, although staffing roles were observed at the encounter level, the dataset did not fully specify which team members were simultaneously present during each encounter, limiting our ability to isolate the causal contribution of RN support to provider efficiency. Future work incorporating detailed team composition at the encounter level would enable more precise assessment of each role.

Additionally, the analysis did not include workplace-level scheduling preferences, regulatory constraints, or insurance-related reimbursement differences, all of which may influence feasible scheduling decisions and care delivery patterns. Integrating this information would support richer constraint formulations in the optimization model and enable the use of more complex generalized linear models to improve interoperability across operational, clinical, and administrative systems. Finally, extending the analysis to a longer observation window would allow evaluation of temporal stability, seasonality, and downstream patient outcomes, strengthening the generalizability of the findings.

Overall, this study demonstrates how OR and

machine learning methods can provide actionable operational insights for Street Medicine. Our analysis identifies the staffing configurations that maximize provider productivity, develops an early warning system for high utilizers, and produces robust scheduling recommendations that ensure consistent care delivery.

## References

- Leo Breiman. 2001. Random forests. *Machine learning*, 45(1):5–32.
- Tianqi Chen and Carlos Guestrin. 2016a. [Xgboost: A scalable tree boosting system](#). In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, page 785–794. ACM.
- Tianqi Chen and Carlos Guestrin. 2016b. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.
- Brett J Feldman, Jeniffer S Kim, Laura Mosqueda, Hurren Vongsachang, Josh Banerjee, Charles E Coffey Jr, Brad Spellberg, Michael Hochman, and Jehni Robinson. 2021. From the hospital to the streets: Bringing care to the unsheltered homeless in los angeles. In *Healthcare*, volume 9, page 100557. Elsevier.
- Rebekah A Kaufman, Mahwish Mallick, Jarvis Thanex Louis, Mollie Williams, and Nancy Oriol. 2024. The role of street medicine and mobile clinics for persons experiencing homelessness: a scoping review. *International journal of environmental research and public health*, 21(6):760.
- Dacre R. T. Knight and 1 others. 2023. Artificial intelligence for patient scheduling in the real-world health care setting: A metanarrative review. *Health Policy and Technology*, 12(4):100824.
- Scott Lundberg and Su-In Lee. 2017. [A unified approach to interpreting model predictions](#). *Preprint*, arXiv:1705.07874.
- Kimberly A Lynch, Taylor Harris, Sachin H Jain, and Michael Hochman. 2022. The case for mobile “street medicine” for patients experiencing homelessness. *Journal of General Internal Medicine*, 37(15):3999–4001.
- Teresa Medellin, Leticia R Moczygemba, and Whitney Thurman. 2024. A qualitative study to describe the nature and scope of street medicine programs in the united states. *International Journal of Environmental Research and Public Health*, 21(12):1623.
- Robert Tibshirani. 1996. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288.

## A Dataset Variables

| Category                       | Variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Team-Level Predictors</b>   | <i>Operational variables:</i> <ul style="list-style-type: none"> <li>• Team location (e.g., CD1, ECM, COLA)</li> <li>• Average commute distance</li> <li>• Number of scheduled visits per day</li> <li>• Time of visit (morning vs. afternoon)</li> <li>• No-show rates</li> <li>• Average provider visit duration</li> </ul> <i>Patient complexity variables:</i> <ul style="list-style-type: none"> <li>• E/M codes</li> <li>• Diagnoses</li> </ul>                                                                                     |
| <b>Staffing Mix Variables</b>  | <i>Staffing-related predictors:</i> <ul style="list-style-type: none"> <li>• Provider type (MD, DO, PA, NP, RN, CHW)</li> <li>• RN coverage level per team</li> <li>• Count of RN visits</li> <li>• Count of provider visits</li> <li>• Number of unique patients seen by each staff type</li> <li>• Total touches (RN + provider)</li> </ul> <i>Staffing-related outcomes:</i> <ul style="list-style-type: none"> <li>• Provider visit volume</li> <li>• Unique patient reach</li> <li>• Workload shifts caused by RN support</li> </ul> |
| <b>Patient-Level Variables</b> | <ul style="list-style-type: none"> <li>• Patient ID</li> <li>• Total visits per month</li> <li>• Provider visits per month</li> <li>• RN or CHW visits</li> <li>• Appointment duration category (20, 30, 60 minutes)</li> <li>• No-show frequency</li> <li>• Diagnosis or E/M code complexity</li> <li>• Location of visit</li> </ul> <i>Outcome:</i> <ul style="list-style-type: none"> <li>• High-utilizer status (binary)</li> </ul>                                                                                                   |
| <b>Scheduling Variables</b>    | <ul style="list-style-type: none"> <li>• Scheduled appointment time</li> <li>• Commute time or distance between visits</li> <li>• No-show probability</li> <li>• Expected visit duration (20, 30, 60 minutes)</li> </ul>                                                                                                                                                                                                                                                                                                                  |

Table 2: Summary of Variables Used Across Productivity Modeling, High-Utilizer Prediction, and Scheduling Optimization.

## B Plot Summary

Figure 3 presents the global SHAP summary plot, ranking features by their mean absolute contribution to the model output across all observations. This figure is provided for reference to complement the feature-level discussions in the main text.

## C Sanity Check

Figure 4 shows sanity checks for key aggregated variables used in the productivity analysis. These plots verify expected monotonic relationships between workload-related features and their corresponding SHAP values, serving as validation of the feature construction and aggregation pipeline.



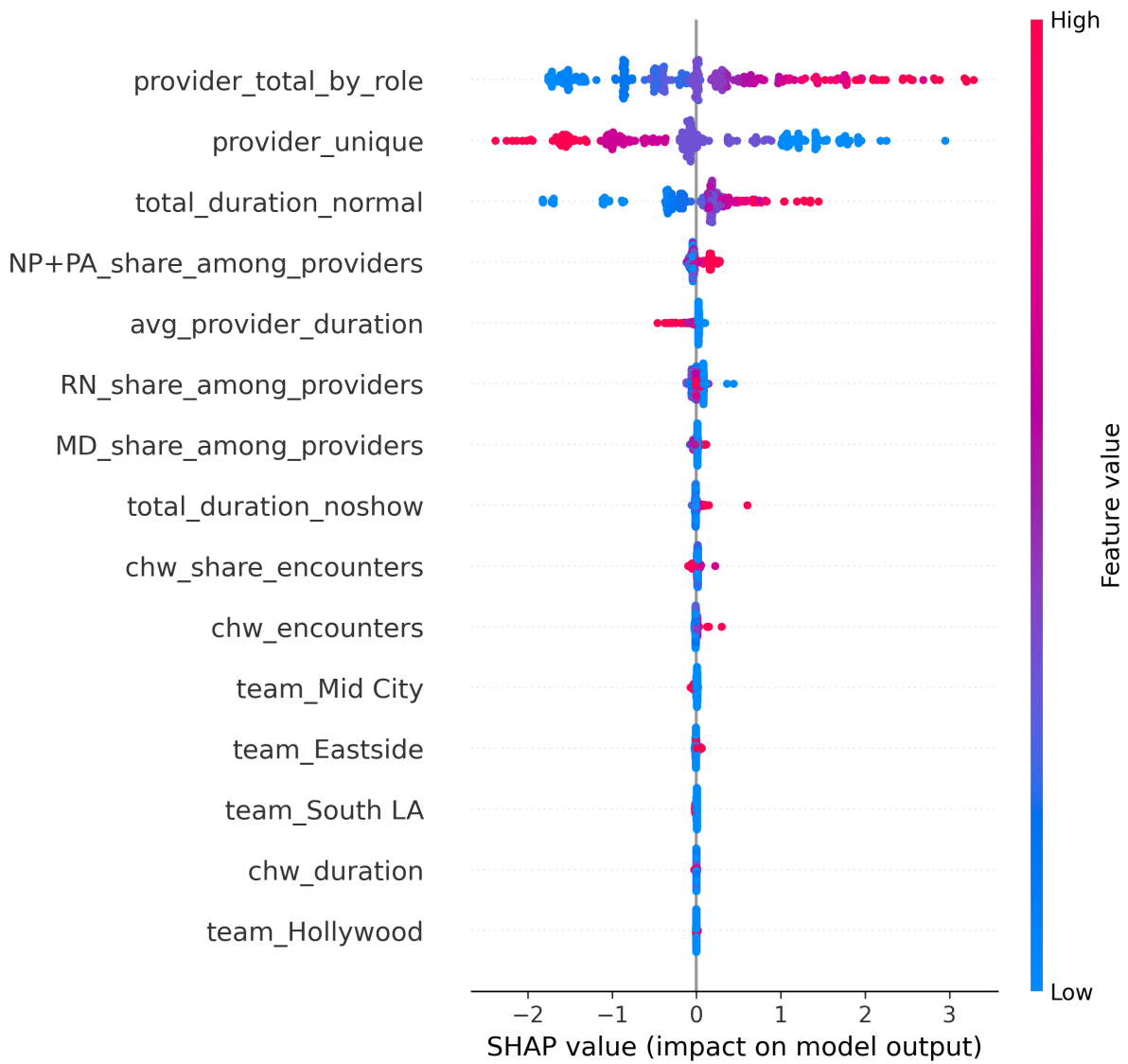


Figure 3: SHAP summary plot for the final XGBoost model.

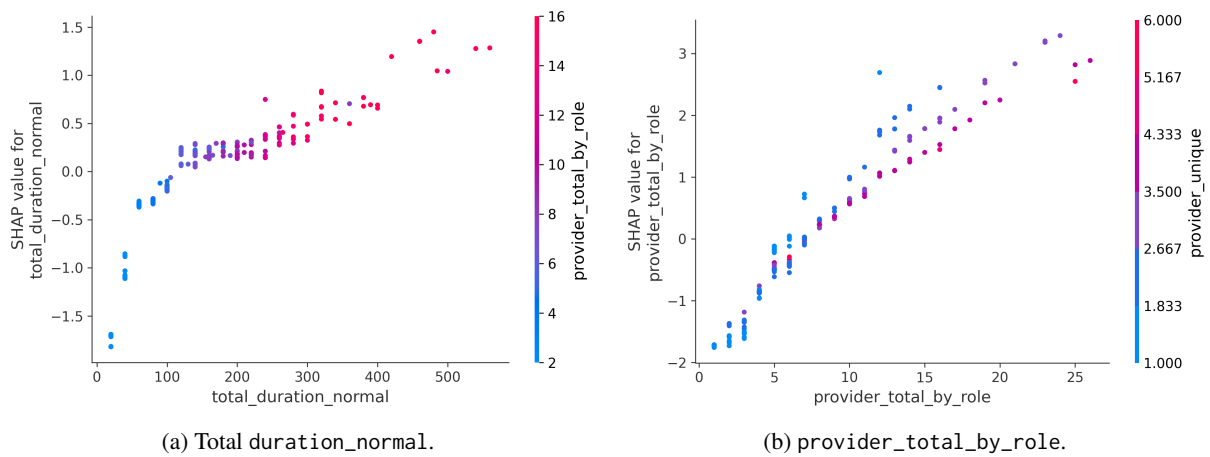


Figure 4: Sanity checks for duration aggregation and provider role totals.